Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Phenomenology of the Standard Model of Elementary Particles

Master's Degree in Physics - Elementary Particles and Astroparticles

Nicolò Favagrossa (68405A)

June 7, 2026

UNIVERSITÀ DEGLI STUDI DI MILANO

Table of Contents

Probing the EW Sector and BSM Physics with $W^{\pm}Zjj$

1 Introduction

- **Dataset:** $140\,\mathrm{fb}^{-1}$ of pp collisions at $\sqrt{s} = 13$ TeV (2015–2018).
- **Target:** Fully leptonic decays of $W^{\pm}Zjj$ production.
- **Measurements:** Integrated & differential fiducial cross-sections (σ_{EW} and σ_{strong}).
- **Goal:** Isolate Vector Boson Scattering (VBS).
- **New Physics:** EFT limits on Anomalous Quartic Gauge Couplings (aQGC).

The ATLAS Detector

1 Introduction

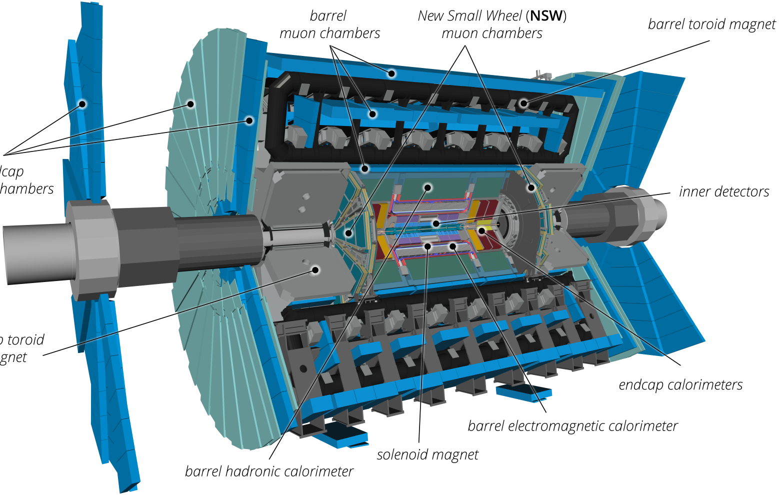


Table of Contents

Gauge Boson Self-Interactions

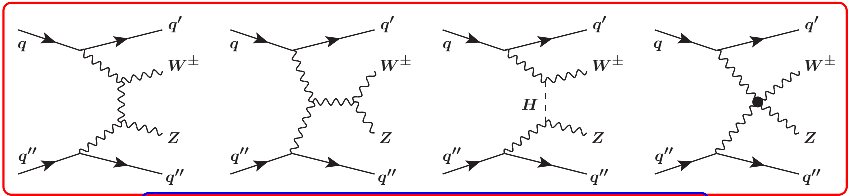
- The Electroweak (EW) sector of the Standard Model is based on the **non-abelian** gauge symmetry $SU(2)_L \times U(1)_Y$.
- Unlike QED, the generators of $SU(2)_L$ do not commute. As a consequence, the physical vector bosons $(W^\pm, Z, \gamma)$ self-interact.
- This structure directly dictates the existence of **Triple** (TGC) and **Quartic** (QGC) Gauge Couplings:

$$\mathcal{L}_{\text{TGC}} = \frac{1}{2} g \epsilon_{abc} W_b^\mu W_c^\nu (\partial_\mu W_{a\nu} - \partial_\nu W_{a\mu}) \qquad \mathcal{L}_{\text{QGC}} = \frac{1}{4} g^2 \epsilon_{abc} \epsilon_{ars} W_b^\mu W_c^\nu W_{r\mu} W_{s\nu}$$

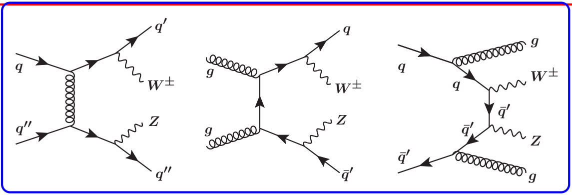
Probe these vertices to test the SM rigidity and search for anomalous couplings (aQGC) arising from New Physics.

Experimental Signature: Vector Boson Scattering

Vector Boson Scattering (VBS) $VV \to VV$ with $V = W/Z/\gamma$, is a key process with which to probe the $SU(2)_L \times U(1)_Y$ gauge symmetry of the **electroweak** (EW) theory that determines the **self-couplings** of vector bosons. To study these couplings, we exploit the purely electroweak production of $W^{\pm}Zjj$.



$W^{\pm}Zjj$**-EW**



$W^{\pm}Zjj$**-QCD**

A little look at the Effective Field Theory (EFT)

The $W^{\pm}Zjj$-EW production can be sensitive to effects beyond the SM affecting the quartic interactions of weak bosons. The measured $W^{\pm}Zjj$ events are therefore used to search for anomalous quartic gauge couplings (aQGC) using an EFT framework. In this model, the Lagrangian of the standard model is extended into an efficient Lagrangian by adding higher-order operators and their respective Wilson coefficients as follows:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i^{(6)}}{\Lambda^2} O_i^{(6)} + \sum_j \frac{f_j^{(8)}}{\Lambda^4} O_j^{(8)} + \dots$$

Where:

- $O_{i,j}^{(6),(8)}$: are dimension-6 and dimension-8 operators;
- $f_i^{(6)}$ e $f_j^{(8)}$: dimensionless couplings
- Λ: is the energy scale of the new processes

A little look at the Effective Field Theory (EFT)

The squared scattering amplitude of the effective field theory predicion for $W^{\pm}Zjj$ production can be written as:

$$\left| A_{\mathrm{SM}} + \sum_i c_i A_i \right|^2 = |A_{\mathrm{SM}}|^2 + \sum_i c_i \, 2\mathrm{Re}(A_{\mathrm{SM}}^* A_i) + \sum_i c_i^2 |A_i|^2 + \sum_{ij, i \neq j} c_i c_j \, 2\mathrm{Re}(A_i A_j^*)$$

- $c_i = f_j^{(8)} / \Lambda^4$, A_{SM} is the SM scattering amplitude;
- $2\,\mathrm{Re}(A_{\mathrm{SM}}^* A_i)$ is the amplitude of the interference term between the SM and the dimension-8 operators;
- $\sum_i {c_i}^2 |A_i|^2$ is the pure dimension-8 contribution, and $\sum_{ij, i \neq j} c_i c_j 2\,\mathrm{Re}(A_i A_j^*)$ is the amplitude of interferences between two dimension-8 operators, called cross terms.

Table of Contents

Fiducial Phase Space for Leptons

The $W^{\pm}Zjj$ cross-sections are measured in a fiducial phase space. Leptons produced in the decay of a hadron, a τ-lepton, or their descendants are not considered in the definition of the fiducial phase space and are treated as background.

Category	Kinematic Variable	Requirement
Z leptons	Transverse momentum ($p_T^{\ell_Z}$)	> 15 GeV
W lepton	Transverse momentum ($p_T^{\ell_W}$)	> 20 GeV
All leptons	Pseudorapidity ($\lvert\eta_\ell\rvert$)	< 2.5
Z boson	Invariant mass window ($\lvert m_{\ell^+\ell^-} - m_Z^{\text{PDG}}\rvert$)	< 10 GeV
W boson	Transverse mass (m_T^W)	> 30 GeV
Angular isolation	Distance between Z leptons ($\Delta R(\ell_{Z1}, \ell_{Z2})$)	> 0.2
Angular isolation	Distance between W and Z leptons ($\Delta R(\ell_W, \ell_Z)$)	> 0.3

Fiducial Phase Space for Jets

In addition, at least two jets are required, reconstructed using the anti-k_t algorithm with radius parameter $R = 0.4$. Hard processes with a b-quark in the initial or final states, such as tZj production, are not considered as part of the $W^{\pm}Zjj$-EW events.

Category	**Kinematic Variable**	**Requirement**
All jets	Transverse momentum (p_T^j)	> 40 GeV
All jets	Pseudorapidity ($\|\eta_j\|$)	< 4.5
Angular isolation	Distance between jets and leptons ($\Delta R(j, \ell)$)	> 0.3
Tagging jets	Opposite hemispheres ($\eta_{j1} \cdot \eta_{j2}$)	< 0
Tagging jets	Invariant mass (m_{jj})	> 500 GeV
	Jet Veto	
b-quark jets	Absence of b-jets in initial/final state	$N_{b\text{-jet}} = 0$

How to eliminate leptons ambiguity?

- Leptons and final-state neutronos that do not originate from hadron or τ-lepton decays, are matched to the $W^{\pm}$ and Z boson decay products using an algorithmic approach, called the **resonant shape algorithm**.
- This algorithm is based on the value of an estimator expressing the product of the nominal line-shapes of the W and Z resonances:

$$P = \left| \frac{1}{m^2_{(\ell^+,\ell^-)} - \left(m_Z^{\mathrm{PDG}}\right)^2 + \mathrm{i}\ \Gamma_Z^{\mathrm{PDG}}\ m_Z^{\mathrm{PDG}}} \right|^2 \times \left| \frac{1}{m^2_{(\ell',\nu_{\ell'})} - \left(m_W^{\mathrm{PDG}}\right)^2 + \mathrm{i}\ \Gamma_W^{\mathrm{PDG}}\ m_W^{\mathrm{PDG}}} \right|^2$$

Monte Carlo Simulation of the Signal

The simulation of $W^{\pm}Zjj$ events was primarily generated using the `MadGraph5_aMC@NLO2.6.5` MC event generator, interfaced with `Pythia8.240`. The simulation was divided into distinct components:

- **Electroweak production** $W^{\pm}Zjj-$ **EW**: The actual signal. It is calculated at leading order (LO), considering purely electroweak processes (order α_{EW}^{6}).
- **QCD production** $W^{\pm}Zjj-$ **QCD**: Considered a background for the VBS analysis. This component is calculated at next-to-leading order (NLO) in QCD.
- **Interference** $W^{\pm}Zjj$**-INT**: Quantifies the quantum interference between the EW and QCD production. It is calculated at LO and yields a positive contribution of 6% in the fiducial region, reaching up to 12% for events with three or more jets.

Monte Carlo Simulation of the Background

In addition to the $W^{\pm}Zjj-$ QCD production, there are numerous other Standard Model processes that can mimic the signal in the detector. These backgrounds fall into two categories based on how they are estimated:

- **Backgrounds with *prompt* (real) leptons:** Processes such as the production of ZZ pairs, WW, tribosons (VVV), or top quarks associated with vector bosons ($t\bar{t}V$, tZj). Their contribution is evaluated entirely using Monte Carlo simulations.
- **Backgrounds with *fake* leptons:** Events where a hadronic jet is misidentified as a lepton by the detector. Being an instrumental artifact that is complex to simulate accurately, this background is estimated directly using data-driven techniques.

Detector Simulation

All generated MC events were passed throught the ATLAS detector simulation, based om `Geant4`, and processed using the same recostruction software used for the data. Furthermore, the lepton and jet momentum scale and resolution, the lepton reconstruction, identification, isolation and trigger efficiencies and the pile-up jets and b-jet veto efficiencies in the simulation were corrected to match those measured in data.

Table of Contents

Event Selection: Physics Objects Definition

Data Quality & Trigger:

- Only data with stable beams and fully operational sub-detectors.
- Single-lepton triggers used (Combined efficiency $\approx 99\%$ for offline-selected $W^{\pm}Zjj$ events).

Physics Objects Reconstruction:

- **Prompt Leptons** $e, \mu, \nu \ (E_T^{miss})$: Reconstructed combining ID tracks with ECAL clusters or MS tracks. Strict isolation and impact parameter criteria are applied to reject non-prompt background.
- **Hadronic Jets:** Clustered using the *anti-*k_t algorithm ($R = 0.4$).
- **b-tagging:** Deep-learning algorithms identify displaced vertices from b-hadron decays (crucial for top-quark veto).

Physics Objects: Leptons

Object	Selection Criteria
Muons	*Medium* Identification (Inner Detector + Muon Spectrometer) $p_T > 15\,\text{GeV}$, $\lvert\eta\rvert < 2.5$ Impact parameters: $\lvert d_0/\sigma_{d_0}\rvert < 3.0$, $\lvert z_0 \sin\theta\rvert < 0.5\,\text{mm}$ Isolation efficiency: $90 - 99\%$ (depending on p_T)
Electrons	*Medium* Likelihood Identification $p_T > 15\,\text{GeV}$, $\lvert\eta\rvert < 1.37$ or $1.52 < \lvert\eta\rvert < 2.47$ (crack region veto) Impact parameters: $\lvert d_0/\sigma_{d_0}\rvert < 5.0$, $\lvert z_0 \sin\theta\rvert < 0.5\,\text{mm}$ Isolation efficiency: $89 - 99\%$ (depending on p_T)

*Note: Impact parameter cuts are crucial to select **prompt** leptons and reject non-prompt background from heavy-flavor decays.*

Physics Objects: Jets & $E_{\mathrm{T}}^{\mathrm{miss}}$

Object	**Selection Criteria**
Jets	Clustered via *anti-k_t* algorithm ($R = 0.4$) $p_T > 25\,\mathrm{GeV}$, $\lvert\eta\rvert < 4.5$ Pile-up suppression:*tight* ($p_T < 60\,\mathrm{GeV}$, $\lvert\eta\rvert < 2.4$) and fJVT *loose* ($p_T < 120\,\mathrm{GeV}, 2.5 < \lvert\eta\rvert < 4.5$)
Missing Energy ($E_{\mathrm{T}}^{\mathrm{miss}}$)	Proxy for the undetected neutrino. Computed as the negative vector sum of all calibrated *hard* objects (leptons, jets) plus a track-based *soft* term.

Event Selection: Boson Reconstruction

Category	Selection Criteria
Baseline	Exactly 3 lepton candidates (μ or e) Veto on a 4th *loose* lepton ($p_T > 5\,\mathrm{GeV}$) Trigger matching: at least 1 lepton with $p_T > 25/27\,\mathrm{GeV}$
Z **Boson Candidate**	Opposite-Sign Same-Flavor (OSSF) lepton pair $\lvert m_{\ell\ell} - m_Z^{\mathrm{PDG}} \rvert < 10\,\mathrm{GeV}$ (If ambiguous, the pair closest to m_Z is chosen)
W **Boson Candidate**	The remaining 3rd lepton + $E_{\mathrm{T}}^{\mathrm{miss}}$ Stricter requirements: *Tight* ID & Isolation, $p_T > 20\,\mathrm{GeV}$ Transverse mass: $m_T^W > 30\,\mathrm{GeV}$

Signal Region (SR): VBS Topology

Category	Selection Criteria
Tagging Jets	The two highest-p_T jets in the event ($p_T > 40\,\mathrm{GeV}$) Must reside in opposite hemispheres ($\eta_{j1} \cdot \eta_{j2} < 0$)
Kinematic Cuts	$m_{jj} > 150\,\mathrm{GeV}$ (rejects triboson background) $m_{jj} > 500\,\mathrm{GeV}$ (isolates the EWK VBS process from QCD)
Background Veto	**Exact b-veto** ($N_{b\text{-jet}} = 0$) (strongly suppresses top-quark background)

Background Estimation Strategy

Two Main Categories:

- **Irreducible:** True prompt leptons mimicking the signal topology.
- **Reducible (Fakes):** Non-prompt leptons or hadronic jets misidentified as prompt leptons.

Estimation Techniques in the Simultaneous Fit:

- **Data-driven (Control Regions):** Used for dominant backgrounds. Extract normalization factors (μ_{bkg}) directly from data.
- **Pure Monte Carlo:** Used for rare processes. Constrained by theoretical uncertainties.
- **Matrix Method:** Fully data-driven approach for reducible fakes, bypassing MC fragmentation modelling.

Irreducible Backgrounds

Constrained via Control Regions (CRs):

- **$ZZ \to 4\ell$ ($ZZjj$-QCD):** Survives via lepton acceptance loss. Constrained in the orthogonal **ZZ-CR** (requiring a 4th *loose* lepton).
- **Top-quark ($t\bar{t} + V, tZj$):** Survives via b-tagging inefficiency. Constrained in the **b-CR** (inverting the b-veto, ≥ 1 b-jet).

Estimated via Pure Monte Carlo:

- **Tribosons (VVV) & $ZZjj$-EW:** Yields too low to define a dedicated CR.
- Handled as Gaussian *Nuisance Parameters* (θ) in the Profile Likelihood Ratio.

Reducible Backgrounds (Fake Leptons)

Sources:

- $Z+$ jets, $Z\gamma$, $t\bar{t}$ without a true 3rd prompt lepton.
- Light/heavy-flavor jets decaying semi-leptonically or photon conversions.

The Matrix Method:

- Relies on two lepton identification criteria: *Loose* (baseline) and *Tight* (signal).
- Uses real lepton efficiency (ϵ) and fake rate (f) measured in independent data samples.
- Solves a linear system to estimate the fake contribution directly in the Signal Region.

Expected and Observed Numbers of Events

	SR ($N_{\text{jet}} = 2$)	SR ($N_{\text{jet}} \geq 3$)	b-CR	ZZ-CR
Data	169	477	666	210
Total pred.	231 ± 12	550 ± 50	660 ± 40	205 ± 11
$W^{\pm}Zjj-$ EW	65.0 ± 3.5	60 ± 6	4.82 ± 0.28	0.725 ± 0.014
$W^{\pm}Zjj-$ QCD	125 ± 9	380 ± 50	77 ± 18	6.2 ± 0.7
$W^{\pm}Zjj$-INT	1.3 ± 0.6	5.3 ± 2.6	0.58 ± 0.29	0.22 ± 0.11
$t\bar{t} + V$	0.66 ± 0.04	20.2 ± 0.7	289 ± 10	9.89 ± 0.28
tZj	8.78 ± 0.34	19.7 ± 1.2	134 ± 4	0.432 ± 0.005
ZZ-QCD	9.6 ± 0.4	32.0 ± 2.5	10.1 ± 0.6	159 ± 9
ZZ-EW	2.2 ± 0.6	4.4 ± 1.1	0.25 ± 0.06	23 ± 6
VVV	0.41 ± 0.10	2.0 ± 0.5	0.39 ± 0.10	4.1 ± 1.1
Misid. leptons	18 ± 4	28 ± 7	150 ± 40	1.7 ± 0.5

Table of Contents

Signal Extraction: Multivariate Analysis (BDT)

The Challenge:

- *WZjj*-EW (signal, $\mathcal{O}(\alpha_{EW}^6)$) and *WZjj*-QCD (background, $\mathcal{O}(\alpha_s^2\alpha_{EW}^4)$) share the exact same final state.
- Cut-based analysis is sub-optimal $\Rightarrow$ Boosted Decision Tree (TMVA).

BDT Input Variables (15 observables):

- *Jet Kinematics:* m_{jj}, $\Delta\eta_{jj}$, $\Delta\phi_{jj}$, jet multiplicity.
- *Boson Kinematics:* p_T^W, p_T^Z, m_T^{WZ}.
- *Mixed Correlations:* Z-centrality (ζ_{lep}), event balance.

BDT Output:

- **The Score**: A single output that allows you to differentiate the signal.

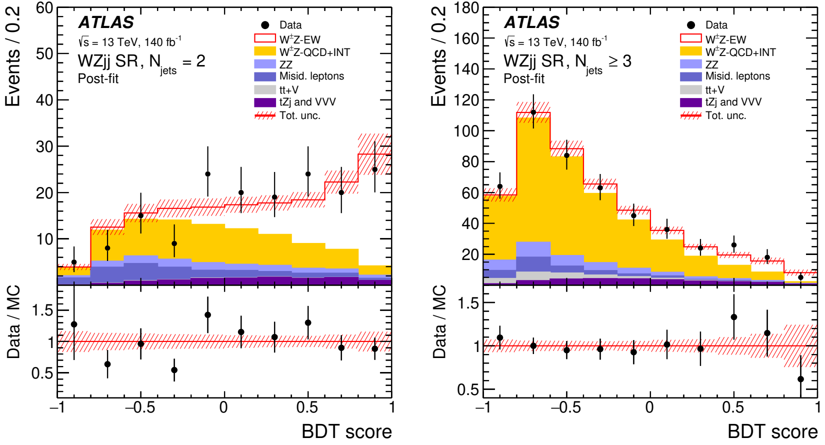
Note: The BDT is also used to differential the background in the b-CR between $t\bar{t}V$ tZj.

Simultaneous Extended Binned Negative Log-Likelihood:

$$-\ln \mathcal{L}(\mu_{sig}, \boldsymbol{K}, \boldsymbol{\theta}) = \sum_r \sum_i \left(\mu^{[r,i]} - N^{[r,i]} \ln \mu^{[r,i]} \right) + \sum_j \frac{\theta_j^2}{2}$$

- **Regions (r):** SR ($N_{jets} = 2$), SR ($N_{jets} \geq 3$), b-CR, ZZ-CR.
- **Observables ($N^{[r,i]}$):** BDT score in SRs, m_{jj} in ZZ-CR, b-CR BDT in b-CR.
- **Parameters of Interest (POI):** Signal strengths μ_{sig} for EWK and QCD.
- **Unconstrained Parameters ($\boldsymbol{K}$):** Background normalizations ($K_{t\bar{t}V}, K_{ZZ}$) extracted dynamically from CRs.
- **Nuisance Parameters (θ):** Theoretical and experimental systematics, constrained by the Gaussian penalty term $\theta_j^2/2$.

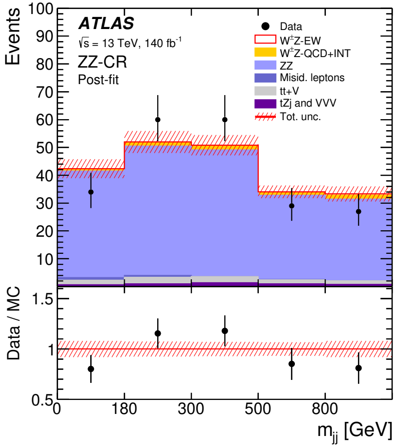
Post-fit distributions SR ($N_{jets} = 2$), SR ($N_{jets} \geq 3$)



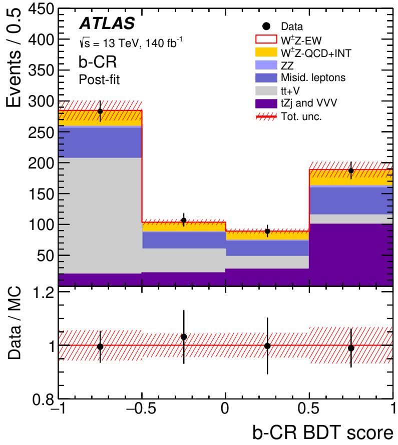
(a) (b)

Post-fit distributions ZZ-CR, b-CR

5 Measurement methodology and results



(c)



(d)

Fiducial Cross-Section Extraction

The fiducial cross-sections are extracted directly from the profile-likelihood fit. The normalisation factors (μ_{EW}, μ_{QCD}) scale the theoretical MC predictions within the fiducial phase space:

$$\sigma_{WZjj-\text{EW}} = \sum_i \mu_i^{WZjj-\text{EW}} \cdot \sigma_{i,th.MC}^{WZjj-\text{EW}}$$

$$\sigma_{WZjj-\text{strong}} = \sum_i \left(\mu_i^{WZjj-\text{QCD}} \cdot \sigma_{i,th.MC}^{WZjj-\text{QCD}} + \mu_i^{WZjj-\text{INT}} \cdot \sigma_{i,th.MC}^{WZjj-\text{INT}} \right)$$

**Index i runs over the two SR categories:* $N_{\text{jets}} = 2$ *and* $N_{\text{jets}} \geq 3$.

- **Interference term:** μ_{INT} is constrained as $\sqrt{\mu_{EW}} \cdot \sqrt{\mu_{QCD}}$.
- **Categorization:** Splitting by N_{jets} minimizes the dependency on the MC modelling of QCD radiation.

Results and Comparison with SM Predictions

VBS (EWK Production)

$$\sigma_{\text{obs}}^{\text{EWK}} = 0.368 \pm 0.037_{\text{stat}} \pm 0.059_{\text{syst}}\ \text{fb}$$

$$\sigma_{\text{th}}^{\text{EWK}} = 0.370 \pm 0.030\,\text{fb}$$

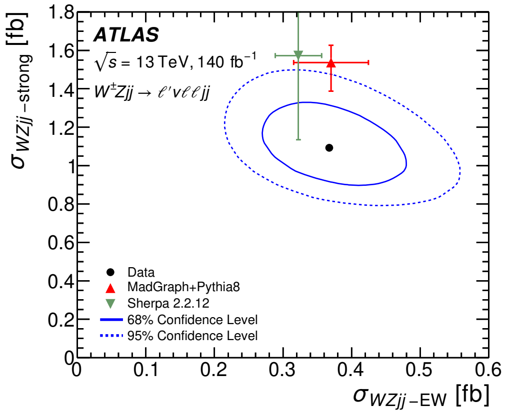
$\Rightarrow$ Excellent SM agreement

Strong + Int.

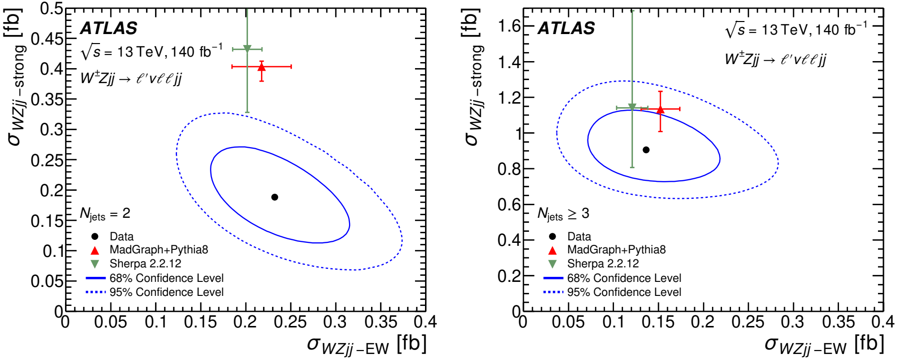
$$\sigma_{\text{obs}}^{\text{strong}} = 1.093 \pm 0.066_{\text{stat}} \pm 0.131_{\text{syst}}\ \text{fb}$$

$$\sigma_{\text{th}}^{\text{strong}} = 1.537 \pm 0.150\,\text{fb}$$

$\Rightarrow$ Deficit factor ~ 0.7 (1.8σ tension)

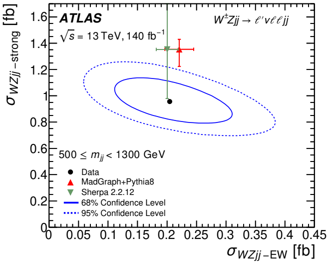


$W^{\pm}Zjj-$ EW and $W^{\pm}Zjj$-strong ($N_{jets} = 2$), SR ($N_{jets} \geq 3$)

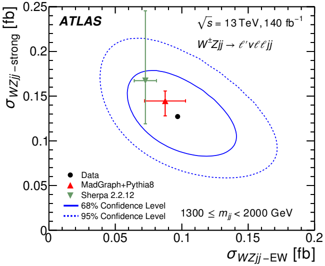


(a) (b)

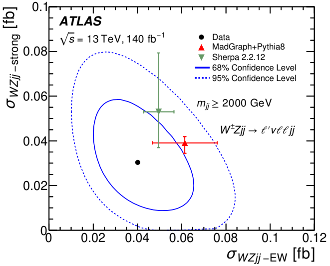
$W^{\pm}Zjj-$ EWand $W^{\pm}Zjj$-strong in differents m_{jj} bins



(a)



(b)



(c)

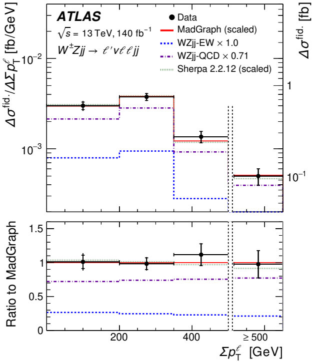
$W^{\pm}Zjj$ differential measurements

Differential mesurements of $W^{\pm}Zjj$ production cross-section are performed as a function of three variables sensitive to anomalies in the quartic gauge coupling in $W^{\pm}Zjj$ events:

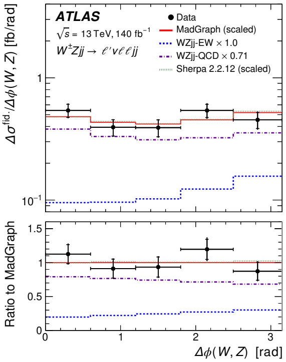
- the scalar sum of the trasverse momenta of the three charged leptons associated with W and Z bosons $\sum p_{\mathrm{T}}^{\ell}$;
- the difference in azimuthal angle $\Delta\phi(W, Z)$ between the W and Z bosons' directions;
- the trasverse mass of the $W^{\pm}Z$ system m_{T}^{WZ}.

$W^{\pm}Zjj$ differential measurements

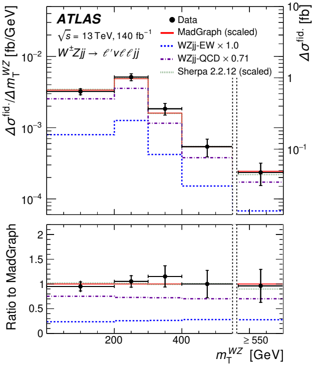
5 Measurement methodology and results



(a)



(b)



(c)

Search for aQGC: Methodology

Anomalous Quartic Gauge Couplings (aQGC) modeled via EFT dimension-8 operators drastically enhance the cross-section at high energy scales ($\hat{s}$).

The 2D Fit Strategy

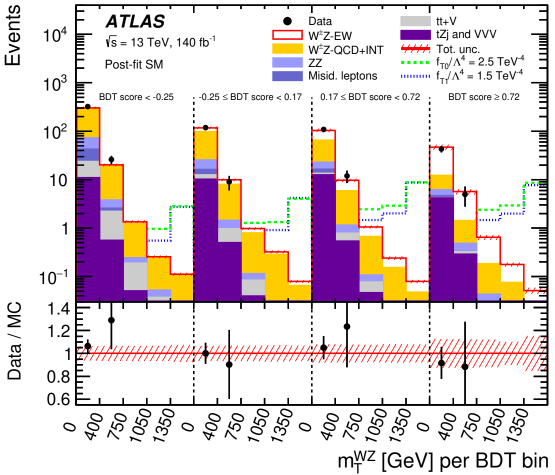
- **Observables:** The search exploits a 2D distribution combining the BDT score (to separate EWK from QCD) and the transverse mass m_T^{WZ} (sensitive to $\hat{s}$).
- **Binning:** 4 Bins in BDT score $\times$ 5 bins in $m_T^{WZ} \to$ Unrolled into a 1D histogram of 20 statistically independent bins.
- **Test Statistic:** Profiled Likelihood Ratio $q(\vec{c})$ to extract 95% Confidence Level (CL) limits on the Wilson coefficients $f_j^{(8)}/\Lambda^4$.

The Matrix Element

The limits are strongly dominated by the pure dimension-8 contribution (quadratic term $c_i^2|A_i|^2$) rather than the SM-EFT interference.

2D BDT vs m_T^{WZ} Distribution for EFT Search

5 Measurement methodology and results



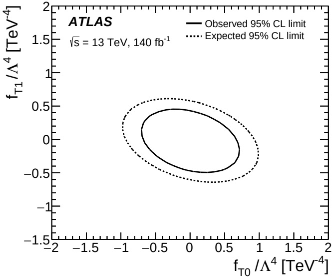
EFT Limits (Without Unitarization)

1D Limits (95% CL)

Coupling	Obs. [TeV^{-4}]	Exp. [TeV^{-4}]
f_{T0}/Λ^4	$[-0.57, 0.56]$	$[-0.80, 0.80]$
f_{T1}/Λ^4	$[-0.39, 0.35]$	$[-0.52, 0.49]$
f_{T2}/Λ^4	$[-1.2, 1.0]$	$[-1.6, 1.4]$
f_{M0}/Λ^4	$[-5.8, 5.6]$	$[-8.3, 8.3]$

- Limits extracted setting one coefficient non-zero at a time.

2D Contour (f_{T0} vs f_{T1})



Tree-Level Unitarity Violation and Clipping

The Unitarity Bound

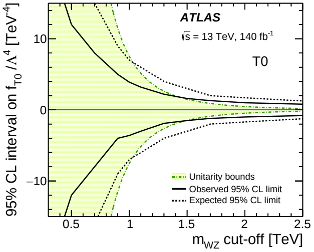
The EFT is not a UV-complete model. Non-zero dimension-8 operators violate tree-level unitarity at high energies.

The "Clipping" Technique

- EFT contributions are removed above a cut-off scale E_c ($m_{WZ} > E_c$).
- Replaced by SM predictions to restore unitarity.
- Limits degrade (widen) as the cut-off E_c becomes more stringent (lower values).

Unitarized Limits (at crossing bound):

- $f_{T0}/\Lambda^4 \in [-1.5, 1.6]$ TeV^{-4}
- $f_{T1}/\Lambda^4 \in [-0.7, 0.6]$ TeV^{-4}



Evolution of 95% CL limits vs cut-off scale

Table of Contents

6 Conclusion

Conclusions: VBS as a Precision Probe for the EWK Sector

From Discovery to Precision Measurement (Run 2, 140 fb^{-1})

- **Integrated cross-section:** $\sigma_{WZjj\text{-EW}} = 0.368 \pm 0.037$ (stat) $\pm\, 0.059$ (syst) fb.
- **Differential measurements** (via Bayesian Unfolding) on aQGC-sensitive variables: $m_{jj}, \Delta\phi, m_T^{WZ}, \sum p_T^{\ell}$.

Standard Model Verification and Modelling Challenges

- **EWK Production:** Excellent Data/Theory agreement in the Signal Region.
- **QCD Background:** Tension of $\mathbf{1.8}\sigma$ (Data $\sim 30\%$ lower than nominal MadGraph).

Limits on New Physics (EFT) and Unitarity

- No high-energy kinematic excesses observed.
- **95% CL limits** set on 9 dimension-8 operators (via Profiled Likelihood Ratio).
- Limits unitarized via theoretical **clipping** to prevent tree-level unitarity violation.

Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Thank you for listening!

Any questions?